

Assignments 2 & 3

— The Workbook



What this is

One repo. Fourteen small exercises. About a month.

You already have the hard part done — VS Code, Git, Node, your first push. This is the JavaScript and web refresher that makes the rest of the semester make sense.

- Each exercise takes **30–45 minutes** and teaches exactly one thing.
- You do them **in order**, a few at a time, on **different days**.
- Everything you need is inside the repo. No videos, nothing to log into.

There is no exam here and nothing to memorise. If you do the work, you pass — and Express, MongoDB and React will feel like more of the same instead of something new.



The loop

The same six steps, fourteen times.

One session, start to finish

Open the next folder and do this:

1. Read its `README.md` — short, one page.
2. Run its `example.js` — *this is the lesson*.
3. Fill in `exercise.js`.
4. Run the tests until they pass.
5. **Commit and push.**
6. Close the laptop. Come back another day.

✓ FOR EXAMPLE, MODULE 04

```
node 04-arrays/example.js
```

Read the comments. Then edit `04-arrays/exercise.js` and:

```
npm test 04
```

```
git add 04-arrays/exercise.js
git commit -m "Finish module 04"
git push
```



It looks like a lot. It isn't.

That loop is six steps the first time and **one habit by the third**.

- You will run it **fourteen times** over the month.
- It is the same loop you will use for **every lab for the rest of the semester** — and in your final project.
- By week two you will stop reading these slides and just do it.

The only way this gets hard is if you save it all for the last week. **Do not accumulate.**
An hour, a few times a week, and you are done comfortably.

Stuck for more than twenty minutes? Ask. That is what we are here for — bring the error message.



Setting up

Three things, once ever.

Get your copy

Open the template: github.com/ng-giu/ebd-w26-workbook

1. Click **Use this template** → **Create a new repository**.
2. Pick any name. Keep it on **your own account**.
3. Clone it — Source Control tab → **Clone Repository**, same as Assignment 1.

✓ THEN, ONCE EVER

Open a terminal in the project (**Terminal** → **New Terminal**) and run:

```
npm install
```

This downloads the tools the exercises use. It takes a minute, and you never do it again.

Then open `START-HERE.md`. It is ten lines.



Submit the form — today, not at the deadline

Fill it in **once**, at the very start: forms.gle/xWbT6eoxfiFLC2eY9

- Your name
- Your student ID
- Your **tutorial number**
- Your **repo's URL**

That is the only thing you ever hand in. After that, **pushing your work is submitting it** — no zip files, no email.



Doing an exercise

Where the teaching actually lives.

Every folder looks the same

File	What it is
README.md	short — what it teaches, what to do
example.js	the lesson — run it, read it
exercise.js	your work — the only file you edit
exercise.test.js	the tests — don't edit

04-arrays/



READ THE EXAMPLE. REALLY.

example.js is not a demo — it **is** the lesson. It runs, it prints, and every line is commented.

The README is one page of orientation. If you read only the README you will be stuck.

Red is not a problem

A failing test tells you exactly what to fix:

```
× returns only the products under 100 EGP  
  → expected undefined to be 45
```

That means your function gave back nothing. Usually a missing `return`.

Work **one module at a time**:

```
npm test 04
```



USEFUL COMMANDS

```
npm test 04      # one module  
npm test         # everything  
npm run modules  # what exists
```

Plain `npm test` will show failures for every module you have not reached yet. **That is normal.** Do not panic — just use `npm test 04`.

Part 2: pages you can see

From module 09 the exercises are web pages, not just JavaScript.

Tests can check your HTML and CSS, but they **cannot see your page**.

Open it and look at it. Every time.

✓ OPEN A PAGE IN YOUR BROWSER

```
npm run open 09          # the example
npm run open 09 exercise # your own page
```

Works on Windows, Mac and Linux.

Press **F12** in the browser for the console — that is where JavaScript errors appear.



The checkpoints

Two of them. These carry the grade.

Your checkpoint is yours

After module 08, and again after module 13:

```
node checkpoint-a/generate.js
```

It reads **your GitHub username** and writes **your** version of the task into `spec.md`.

Different city, different filter, different wording — for every single student.

- The fourteen exercises are marked on **completion**.
- The **checkpoints carry the grade**.
- Copying a friend's answer fails your tests, because their task is not your task.

Nothing new to learn. Everything the spec asks for is in a module you have already done — it just doesn't hand you the function signatures.



Deadlines

What	Due
Assignment 2 Modules 00–08 + Checkpoint A	Fri 10 October
Assignment 3 Modules 09–13 + Checkpoint B	Wed 29 October

No commit created after 31 October is reviewed. At all.

One repo, two grading passes. Push before each date — we grade what is on GitHub.



Commits on six different days

We check that your commits land on at least **six different calendar days** across the month.

This is not extra work. It is the same work, **not crammed into one night** — and spacing it out is the single biggest thing that makes it stick.

Commit whenever a module goes green. That is it.

If something goes wrong

Problem	Fix
<code>npm</code> not recognised	Close the terminal, open a new one. Still failing? Reinstall Node, reopen VS Code.
<code>npm error code ENOENT ... package.json</code>	You are in the wrong folder. Terminal → New Terminal opens the right one.
<code>Could not find vitest</code>	Run <code>npm install</code> from the project's root folder.
Windows: <code>npm.ps1 cannot be loaded</code>	Use Git Bash — the + dropdown in the VS Code terminal panel.
<code>No test files found</code>	Check the number: <code>npm test 04</code> , not <code>npm test 4-arrays</code> .
A value is mysteriously <code>undefined</code>	A missing <code>return</code> , or a missing <code>await</code> .
Page is blank in the browser	Press F12 → Console. An error in your JS stops the whole file.

Each module's `README.md` has more. And ask us.

Questions?

Submit once, today — then just push your work

